

Q6) Calculates charging & parking cost. It checks slot availability battery level & time. Assigns priority based on emergency & prints final cost.

PAC Chart

Given Data
Vehicle, Battery
Duration

Process.
Calculation.

Required Results.
Vehicle, Status
Cost.

Alternative Solution.
Nested loops.

IPO Chart

Input
Vehicle
Disability.
Charge

Process.
Calculation
& Checking
Inputs

Output
Vehicle
Priority.
Amount

INPUT Vehicle, Battery, ChargeLevel, Duration, Time,
Membership, Disability, Availability.

Discount, Charge, cost, ParkCost, FinalCost $\leftarrow 0$.
Priority $\leftarrow ' '$

IF Charging Availability = TRUE.

IF Vehicle = 'H'

PRINT 'Charging Unavailable - parking
only'

ELSE PRINT 'No slot available'

ELSE

IF Type = H AND $\frac{1}{2}$ Battery < 40 :

PRINT 'Vehicle does not qualify
for EV Charge'

ELSE

Charge $\leftarrow$ ChargeLevel - Battery.

IF Charge $\leq$ Battery:

Priority $\leftarrow$ 'Emergency Charging'

ELSE IF membership = 'Y' AND

Battery ≤ 30

priority $\leftarrow$ 'Priority Charging'

ELSE

priority $\leftarrow$ 'Normal Charging'

IF Time ≥ 1700 AND Time ≤ 2200 ,

PRINT 'Peak Time'

ChargeCost $\leftarrow 50 \times \text{Duration}$.

IF Membership = 'Y' AND NOT
Emergency :

Discount $\leftarrow 10$.

ChargeCost $\leftarrow \text{ChargeCost} \times 0.9$

ELSE

PRINT 'OFF-Peak'

ChargeCost $\leftarrow 35 \times \text{Duration}$.

IF Membership = 'Y' AND NOT
Emergency :

Discount $\leftarrow 20$.

ChargeCost $\leftarrow \text{ChargeCost} \times 0.8$.

ENDIF

IF Duration < 2

ParkCost $\leftarrow 200$.

ELSE IF Duration ≥ 2 AND Duration ≤ 5

ParkCost $\leftarrow 500$

ELSE ParkCost $\leftarrow 700$

IF Membership = 'Y'.

Park Cost $\leftarrow$ Park Cost * 0.8.

IF Disability = 'Y'

Park Cost $\leftarrow$ 0.

IF Duration > 8.

PRINT 'Relocate Vehicle After Charging'.

ELSE PRINT 'Standard Parking Location'.

PRINT 'Vehicle', Vehicle.

PRINT 'Battery', Battery.

PRINT 'Required Charging', Charge.

PRINT 'Priority', Priority.

PRINT 'Charge Cost', ChargeCost.

PRINT 'Parking Cost', ParkCost.

PRINT 'Discount', Discount.

PRINT 'Final Cost', ChargeCost + ParkCost.

Start

INPUT Vehicle, Battery,
Charge level, Duration,
Time, Membership, Disability,
Availability

Discount, Charge, ChargeCost,
ParkCost, Final Cost $\leftarrow 0$

IF Availability
= TRUE

No

Vehicle
= H

Yes

PRINT 'Charging
Unavailable'

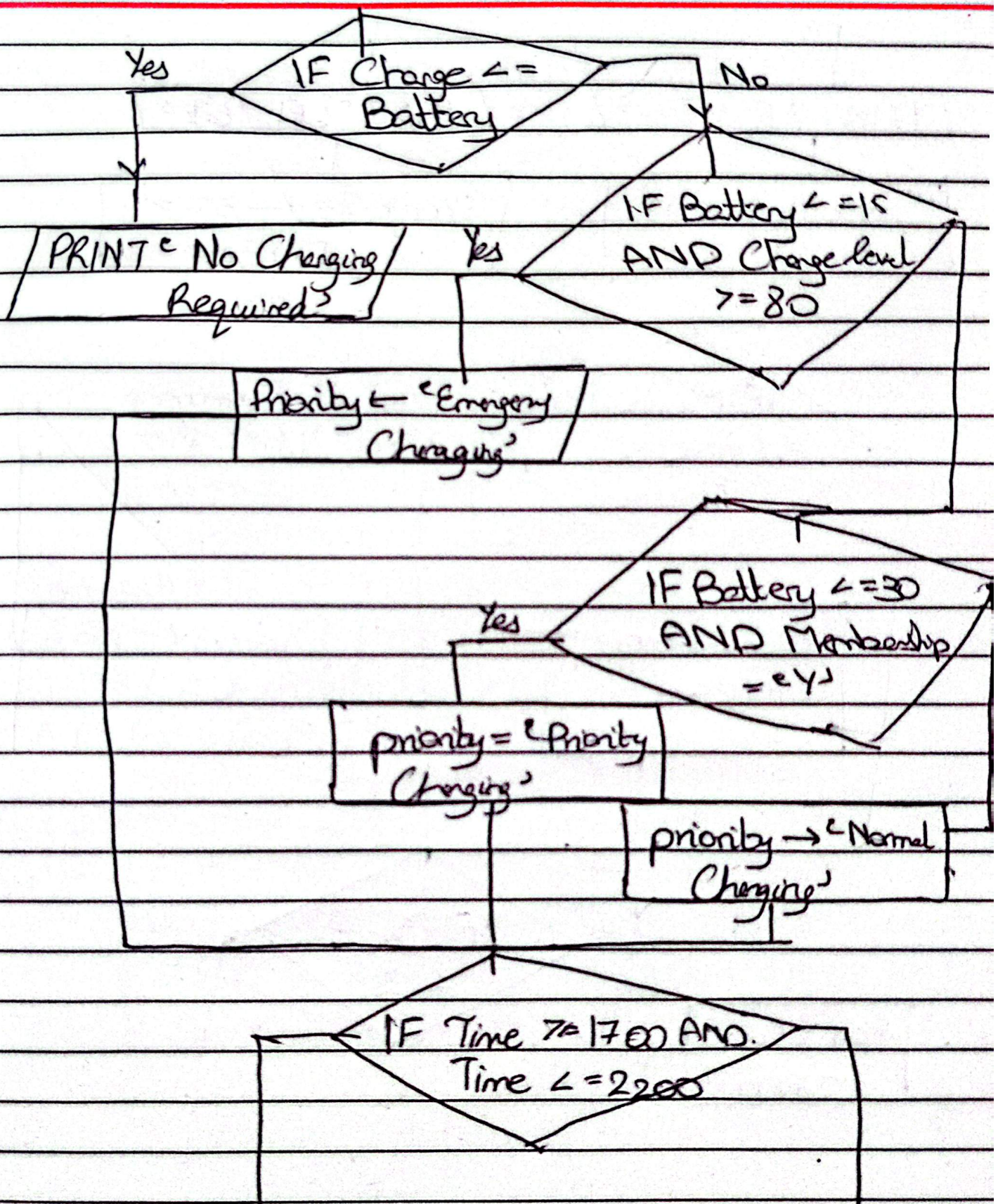
PRINT 'No Charging
Slot Available'

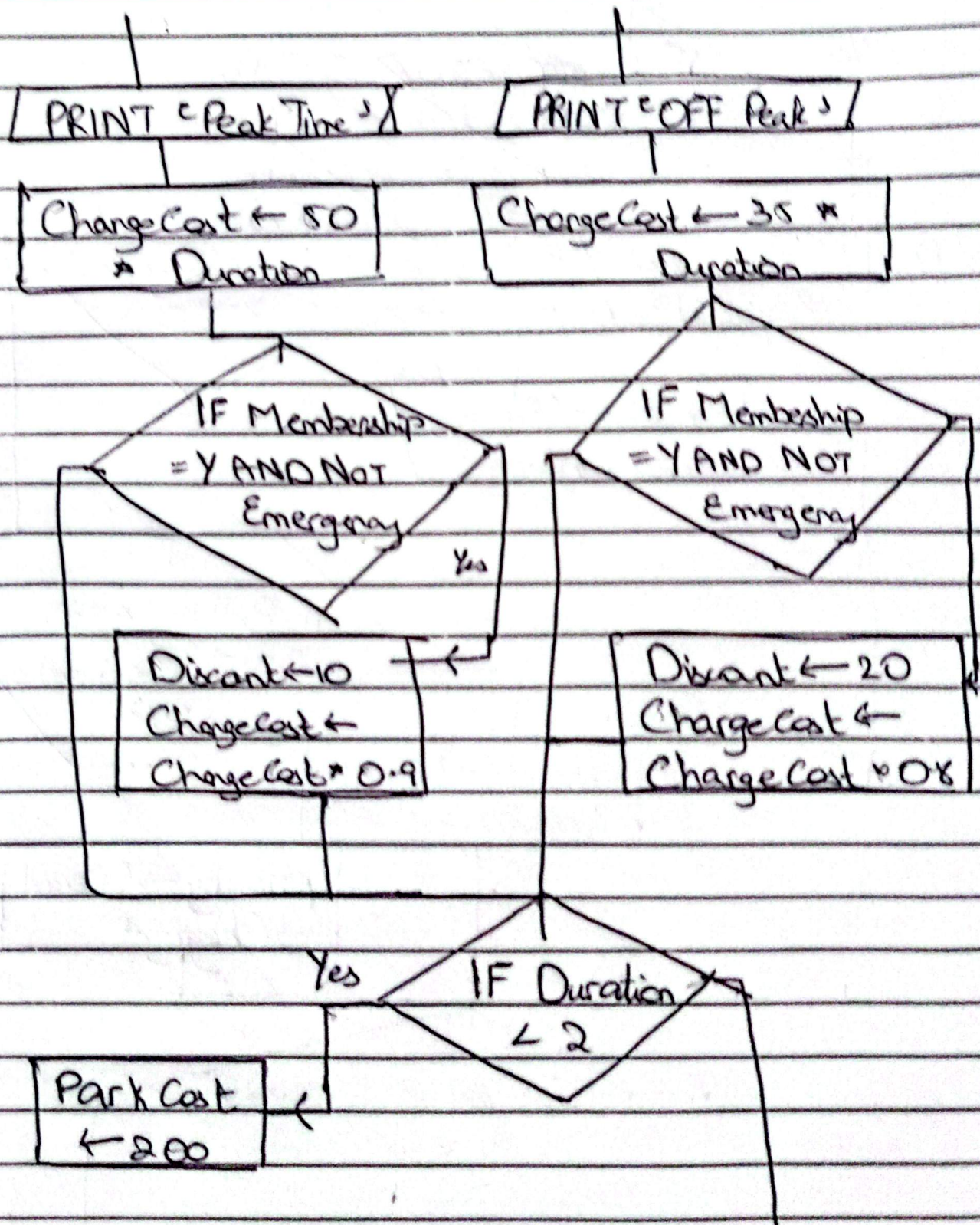
IF
Vehicle = H AND
Battery < 40

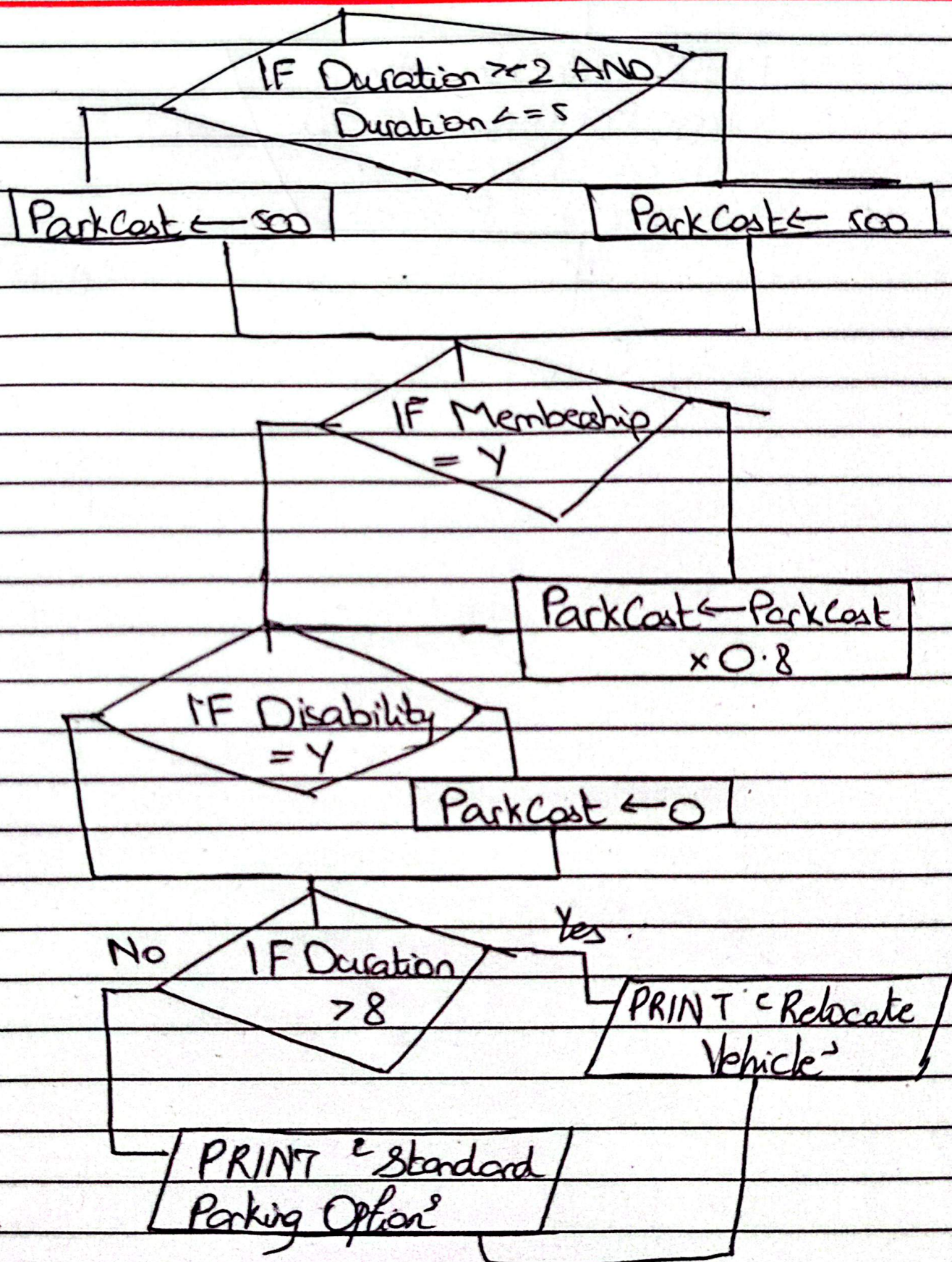
Yes

PRINT 'Vehicle does
not qualify'

Charge = ChargeLevel - Battery







Date: _____

